



Iranian Mathematical Society Competition

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by mojyla222

– Day 1

- P1** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing function. Suppose that for every $a, b \in \mathbb{R}$, the following equality holds:

$$\int_a^{f(b)} f(t) dt = \int_{f(a)}^b f(t) dt.$$

Prove that for every $x \in \mathbb{R}$, $f(x) = x$.

- P2** Let G be a finite group where the order of every non-trivial subgroup is a prime number. Prove the following:

A: G is generated by 2 elements.

B: G is a simple group (a group is simple if it has no non-trivial normal subgroups).

- P3** Let $a_1, a_2, \dots, a_n$ be distinct natural numbers. Show that the set

$$A = \left\{ \frac{a_i}{a_j} \mid 1 \leq i, j \leq n \right\}$$

contains at most $n - 1$ distinct prime numbers.

- P4** Let G be a finite group. Suppose there exists a function $f : G \rightarrow G$ such that for every $x \in G$,

$$f(xa) = xf(x),$$

where $a \in G$ is a fixed element. Prove that the order of G is odd.

- P5** Let (X, d) be a metric space, and let $f : X \rightarrow \mathbb{R}$ be a function with the property that

$$|f(x) - f(y)| \leq d(x, y) + 1 \quad \text{for all } x, y \in X.$$

Prove that there is a function $g : X \rightarrow \mathbb{R}$ such that for all $x, y \in X$:

$$|g(x) - f(x)| \leq 1, \quad |g(x) - g(y)| \leq d(x, y).$$

P6 Let $a_1, a_2, \dots, a_n$ be natural numbers whose greatest common divisor is 1. Prove that there exists an $n \times n$ matrix with integer entries and determinant 1, whose first row is $[a_1 \cdots a_n]$.

– Day 2

P7 Prove that a closed disc in $\mathbb{R}^2$ with radius 1 cannot be covered by two discs each with radius less than 1. (Here, a "disc" refers to a closed ball.)

P8 Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an analytic function such that for every $z \in \mathbb{C}$, $f(z) = f(z - z^2)$. Prove that f is a constant function.

P9 Let A be a square matrix with real entries such that the matrix $A + A^T$ is idempotent. Prove that the sum of all entries of A is non-negative. (A matrix B is called idempotent if $B^2 = B$.)

P10 Prove that there exists a sequence $\alpha_1, \alpha_2, \dots$ of numbers in $(-1, 1)$ such that for every $n \geq 1$, the partial sums

$$S_n = \alpha_1 + \frac{\alpha_2}{2} + \cdots + \frac{\alpha_n}{n}$$

form a dense subset of $\mathbb{R}$.

P11 Consider the space $\mathbb{R}^n$ with the Euclidean metric d . Let x be a random vector in $\mathbb{R}^n$ where each component is independently equal to 1 with probability $\frac{1}{2}$ and equal to -1 with probability $\frac{1}{2}$. Let W be a k -dimensional subspace of $\mathbb{R}^n$. Compute the expected value of $\text{dist}(x, W)^2$, where

$$\text{dist}(x, W) = \inf\{d(x, y) : y \in W\}.$$

P12 Let R be an integral domain such that every non-constant polynomial in $R[x]$ has a root in R . Let P and Q be two prime ideals in R . Prove that for any $a, b \in R$ such that $ab \in P + Q$, either $b \in P + Q$ or $a \in P + Q$.